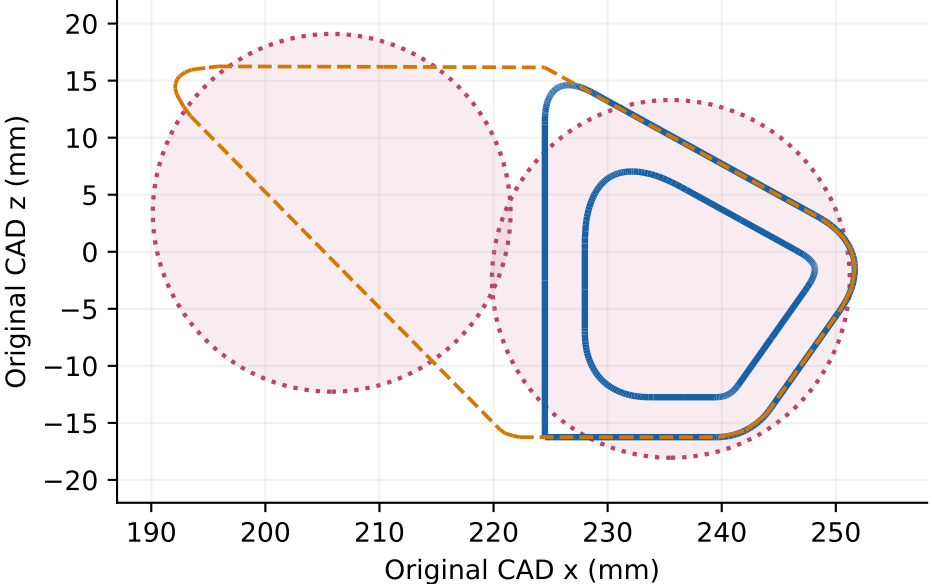


Silicone foot: CAD geometry and collision-envelope comparison

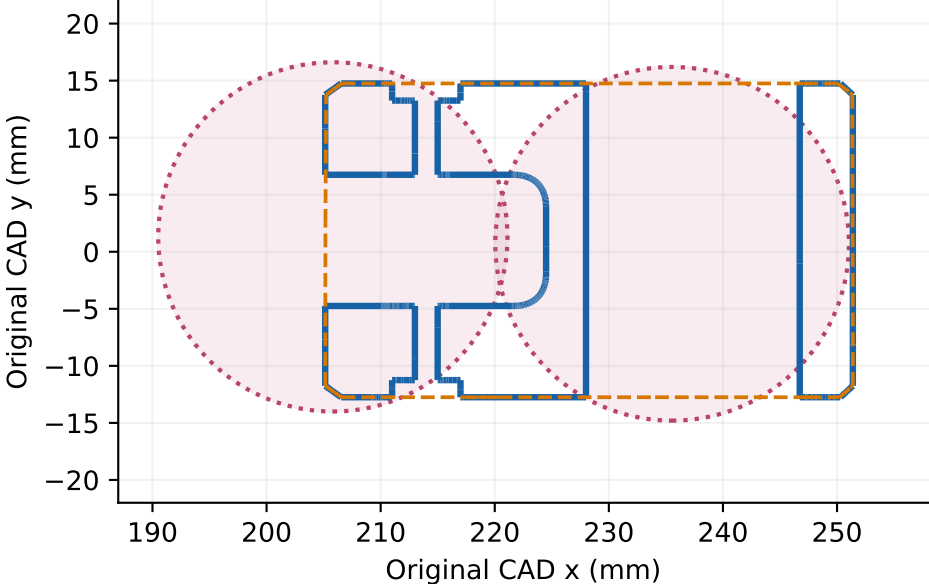
True planar sections, equal axis scales, millimeter dimensions. No mesh recentering or rescaling.

— Exact triangulated CAD section Current two-sphere sections - - - 64-vertex convex candidate

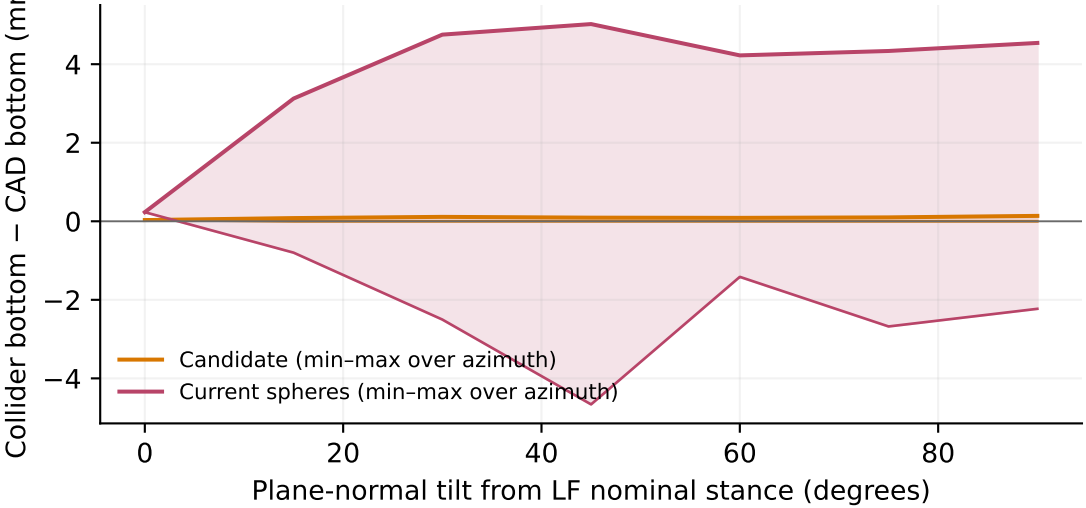
Longitudinal section: CAD y = 1 mm



Orthogonal section: CAD z = 0 mm



Contact-height error across 72 azimuths per tilt



0.158 mm maximum candidate undercoverage
9.816 mm maximum current-sphere undercoverage

64 vertices / 124 triangles • original CAD coordinates
Bounds cover the entire triangulated CAD surface.

7.807 mm sampled cavity overfill: see CAD sections.
Collision volume must not replace the mass model.

UNQUALIFIED RIGID ENVELOPE PROPOSAL

Not installed in the robot simulation.
Silicone compliance and contact behavior unmeasured.

Positive bottom error: collider sits above the CAD surface. Support-height plots do not reveal the two-sphere waist.